

Bilal Ahmad Kalas

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Profile

M.Sc. Mathematics graduate with interests in algebra, topology, number theory, and algebraic structures. My academic work has focused on finite transformation semigroups and independent study in pure mathematics. I am seeking doctoral research in algebraic structures and commutative algebra, particularly where algebra connects with topology, number theory, and geometry.

Education

Integrated B.Sc.-M.Sc. Mathematics
Central University of Kashmir, India

2019–2024

Master's Dissertation

Dissertation: Ranks and Presentation for Finite Transformation Semigroups
Supervisor: Dr Aftab Hussain Shah (Associate Professor)

[Link](#)

- Studied rank and presentation problems for finite order-preserving transformation semigroups.
- Based on the work of Joerg Koppitz and Somuek Worawiset on transformations with one fixed point.
- Focused on semigroup presentations, generating set, ranks, and structural & combinatorial properties of finite transformation semigroups.

Research Interests

Algebra: Commutative Algebra, Field Theory, Galois Theory, Semigroup Theory, Representation Theory; Algebraic Topology, Algebraic Number Theory, Algebraic Geometry, Topology

Teaching Experience

Private Mathematics Tutor

2024-2025

Skills

- Relevant Coursework: Abstract Algebra, Commutative Algebra, Galois and Field Theory, Topology, Number Theory, Semigroup Theory
- Academic Skills: Mathematical proof writing, mathematical exposition, independent study, reading and analysis of mathematical literature.
- Technical skills: Latex, Python, HTML/CSS, JavaScript, C

Languages

- Hindi/Urdu – Native
- Gojri – Native
- Kashmiri – Native
- English – Fluent